

MODULE 22 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Cost Optimisation by Design

Closing Domain 2. The charges that never appear on an architecture diagram.

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Every arrow on your diagram has a price.
Data transfer is the cost nobody draws.

Who pays, and how much

PATH	ROUGHLY	NOTES
Into AWS	Free	Ingress is never charged
Same AZ, private IP	Free	Public IP in the same AZ is charged
Cross-AZ	\$0.01/GB each way	So \$0.02 round trip. The quiet killer
Inter-Region	\$0.01 to \$0.02/GB	Replication doubles a database bill
Out to the internet	~\$0.09/GB	First 100 GB a month is free
Through a NAT gateway	+\$0.045/GB	On top of egress, not instead of

ON THE EXAM

These are US list prices for orientation, not values to memorise. What matters is the shape: cross-AZ is charged both ways, and NAT stacks.

The NAT gateway trap

- ◆ **A NAT gateway charges per hour and per GB processed.** Three for redundancy costs roughly \$100 a month before any traffic.
- ◆ **S3 traffic from a private subnet routed through NAT is billed per GB,** even though S3 in-Region transfer is free.
- ◆ **A gateway endpoint for S3 or DynamoDB is free** and removes that charge entirely. This is the single cheapest fix in AWS.
- ◆ Interface endpoints cost about a penny per GB plus an hourly fee, which still undercuts NAT for AWS service traffic.

ON THE EXAM

Backing up terabytes to S3 through a NAT gateway is the classic. The data transfer is free; the NAT processing is not.

Charges that arrive quietly

- ◆ **Every public IPv4 address costs about \$0.005 an hour since 2024**, attached or idle. Roughly \$3.60 a month each.
- ◆ **Cross-AZ chatter** between microservices is charged both ways, and does not appear on any diagram.
- ◆ **Multi-AZ RDS replication is free** between primary and standby. Cross-Region read replicas are not.
- ◆ **S3 or EC2 to CloudFront origin fetches are free**, which is why serving through CloudFront beats serving directly.

ON THE EXAM

A cluster spread across three AZs with fifty chatty services can generate several TB a month of cross-AZ traffic on its own.

Design levers that actually move the number

- ◆ **Serve through CloudFront.** The origin fetch is free and cached responses never touch your origin again.
- ◆ **Keep chatty components in one AZ,** and accept the availability trade deliberately rather than by accident.
- ◆ **Gateway endpoints for S3 and DynamoDB,** always. There is no reason not to.
- ◆ **Compress and batch.** Halving payload size halves every per-GB charge on its path at once.

ON THE EXAM

Compression is the lever people forget. It reduces egress, cross-AZ, NAT processing and inter-Region charges simultaneously.

Cost decisions made at design time

- ◆ **Managed versus self-managed** is a staff cost question. Compare the instance bill against the people running it.
- ◆ **Serverless is cheapest when idle and dearest when saturated.** At steady high volume, containers usually win.
- ◆ Right-size at design time using load tests, then let Compute Optimizer correct you once real traffic arrives.
- ◆ A commitment made before the architecture settles becomes a commitment to the wrong thing, as in module eleven.

ON THE EXAM

Serverless economics invert somewhere. Find the crossover for the workload rather than assuming it is always cheaper.

Reading the scenario

THE SCENARIO SAYS

Large NAT charges, traffic goes to S3

Network cost exceeds compute cost

Heavy egress serving files to users

Idle Elastic IPs across many accounts

Cross-Region replication is expensive

Steady high-volume Lambda

IT IS POINTING AT

A gateway endpoint

Cross-AZ chatter, or NAT processing

CloudFront

The public IPv4 hourly charge

Ask what RPO actually requires

Compare against Fargate or EC2

ON THE EXAM

Ask which boundary each byte crosses. Most surprise charges are a boundary somebody did not know was there.

Meridian Freight's analytics platform writes 90 TB a month to Amazon S3 from EC2 instances in private subnets. The S3 bucket is in the same Region. Finance has flagged a NAT gateway line of roughly \$4,000 a month, dominated by data processing rather than hours. The instances also need occasional internet access for OS patching.

What should the solutions architect recommend?

- ☐ A Add a gateway endpoint for S3 to the private subnets' route tables
- ☐ B Move the instances to public subnets to bypass the NAT gateway
- ☐ C Consolidate to a single NAT gateway to reduce hourly charges
- ☐ D Compress the data before writing it to S3

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** S3 gateway endpoint
Free, removes 90 TB from the NAT gateway entirely, and the gateway stays for patching traffic. Nothing else changes.
- B** Public subnets
Removes the NAT charge by putting analytics instances directly on the internet. A security regression to solve a billing problem.
- C** One NAT gateway
The bill is dominated by processing, not hours. Consolidating also creates a single point of failure for a saving of about \$65.
- D** Compress the data
Genuinely helps and is worth doing anyway. It halves a charge that a gateway endpoint removes completely, and for free.

THE SIGNAL

Dominated by processing rather than hours is the sentence that decides it. Options C and D attack the wrong term.

A company runs 60 microservices on EKS across three Availability Zones. Services call each other heavily. Cost analysis shows regional data transfer charges of about \$9,000 a month, larger than the EC2 bill for the cluster itself.

Availability requirements permit some services to run in fewer zones.

What should the solutions architect investigate FIRST?

- ☐ A Cross-AZ traffic between services, charged in both directions
- ☐ B Internet egress from the cluster to end users
- ☐ C The cost of the EKS control plane
- ☐ D EBS volume charges for the worker nodes

◆ PAUSE AND DECIDE

What should the solutions architect investigate FIRST?

- A

Cross-AZ service traffic
Sixty services scheduled randomly across three zones means most calls cross a zone boundary and are charged twice, once each way.
- B

Internet egress
Would appear as internet data transfer out, not as regional transfer. The scenario names the line item.
- C

EKS control plane
A fixed hourly charge per cluster. It is not data transfer and does not scale with chatter.
- D

EBS charges
Storage, not transfer. It would appear under EBS rather than under data transfer.

THE SIGNAL

Regional data transfer is the line item, and it names the boundary. Cross-AZ is the only one that grows with service chatter.

A training company serves 400 TB a month of video files directly from Amazon S3 to learners worldwide, using pre-signed URLs. The egress line is the single largest item on the bill. Files are large, popular, and rarely change.

The team believes CloudFront would only add cost.

What should the solutions architect advise?

- ☐ A They are correct; CloudFront adds a layer and a charge
- ☐ B Serve through CloudFront: the S3 origin fetch is free and cached objects are served once from origin
- ☐ C Move the files to S3 Glacier Instant Retrieval to reduce cost
- ☐ D Use S3 Transfer Acceleration for the downloads

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A They are correct

It feels right and is wrong. CloudFront replaces the S3 egress charge rather than adding to it, and its rate is lower at volume.

C Glacier Instant Retrieval

Reduces storage cost for files that are popular, and adds retrieval fees. Storage was never the large line.

B Serve through CloudFront

S3 to CloudFront transfer is free, CloudFront egress undercuts S3 egress, and popular files are fetched from origin once.

D Transfer Acceleration

Speeds long-distance uploads and downloads at an additional per-GB premium. It increases cost rather than reducing it.

THE SIGNAL

Large, popular, rarely changing files is the ideal CDN profile. The origin fetch being free is what makes it a saving rather than a layer.

A cost review across 140 accounts finds 2,300 Elastic IP addresses. About 1,900 are attached to running instances and 400 are unattached. A junior engineer reports that only the 400 unattached addresses cost anything. Finance wants to know the real figure.

What should the solutions architect explain?

- ☐ A The engineer is correct; only unattached Elastic IPs are charged
- ☐ B All public IPv4 addresses have been charged hourly since 2024, so all 2,300 cost money
- ☐ C Elastic IPs are free; only data transfer through them is charged
- ☐ D The charge applies only to addresses in public subnets

◆ PAUSE AND DECIDE

What should the solutions architect explain?

- A

 Only unattached are charged
The rule until February 2024, and widely still believed. AWS extended the charge to every public IPv4 address.
- B

 All 2,300 are charged
About \$0.005 an hour each, roughly \$3.60 a month, so 2,300 addresses cost in the region of \$8,000 a year.
- C

 Free, transfer only
Was never true even before 2024. Unattached addresses have carried an hourly charge for many years.
- D

 Public subnets only
Invented. The charge attaches to the address itself, not to where it is used.

THE SIGNAL

The engineer is quoting a rule that expired in 2024. Worth auditing whether every one of the 1,900 attached addresses is needed.

A team replicates 10 TB of data daily from eu-west-1 to us-east-1 for disaster recovery, costing around \$6,000 a month in inter-Region transfer. On review, the documented RPO for this dataset is 24 hours, and the data is derived and fully reproducible from a source system. The replication was configured three years ago and never revisited.

What should the solutions architect recommend?

- ☐ A Keep replicating; disaster recovery justifies the cost
- ☐ B Match the replication to the actual RPO, and consider regenerating derived data instead of replicating it
- ☐ C Switch to S3 Replication Time Control for a stronger SLA
- ☐ D Compress the data before replication

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Keep replicating
Treats DR as beyond questioning. The requirement is 24 hours and the design delivers far more, at \$72,000 a year.
- B

 Match the RPO, regenerate what is derived
Derived, reproducible data does not need replicating at all. Replicate the source and rebuild after failover.
- C

 Replication Time Control
Adds a stronger SLA and additional per-GB cost, tightening an RPO that is already far ahead of the requirement.
- D

 Compress it
Reduces the bill and keeps replicating data the business does not need replicated. It optimises the wrong decision.

THE SIGNAL

Two planted details: the RPO is 24 hours, and the data is reproducible. Either alone would make this over-engineered.

An image-processing service runs on Lambda, handling a steady 900 million invocations a month at around 3 seconds and 3 GB each. Load is flat around the clock with no idle periods. The Lambda bill is roughly \$95,000 a month.

The team chose serverless because it is cheaper.

What should the solutions architect advise?

- A Keep Lambda; serverless is always cheaper than running servers
- B Model the same workload on Fargate or EC2, since serverless economics invert at sustained saturation
- C Reduce the memory allocation to cut cost
- D Buy a Compute Savings Plan to cover the Lambda spend

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

 Serverless is always cheaper
True when load is spiky or idle. Flat, saturated load is the case where paying per millisecond stops being an advantage.
- B

 Model it on Fargate or EC2
Continuous saturation is what always-on compute is priced for. Model it rather than assume, but the crossover is clearly worth checking.
- C

 Cut the memory
Lambda CPU scales with memory, so less memory usually means longer duration. It often costs more, not less.
- D

 Compute Savings Plan
Genuinely covers Lambda and is worth having. It discounts the wrong architecture rather than questioning it.

THE SIGNAL

Flat load, no idle, sustained saturation. Every property that makes serverless economical is absent from this scenario.

What to carry out of this module

- ◆ **Every arrow has a price.** Ask which boundary each byte crosses and who pays for it.
- ◆ **Cross-AZ is charged both ways**, and NAT processing stacks on top of egress rather than replacing it.
- ◆ **Gateway endpoints are free.** S3 traffic through a NAT gateway is the most common avoidable charge in AWS.
- ◆ **Serverless inverts at saturation.** Flat, always-busy load is where containers usually win.

ON THE EXAM

Every public IPv4 address has been charged hourly since 2024, attached or not. Most estates have never audited this.